

Black Hole Cosmogenesis: the Standard Model and Cosmology with Zero Free Dimensionless Parameters (Paper 22)

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Abstract

We present a cosmogenesis scenario in which a daughter universe nucleates from a near-extremal Kerr black hole and inherits a *no-hair tuple*—its parent’s mass-hair (fixing the Planck scale M_{Pl}), spin axis, substrate algebra, and a chirality sign. From this *single dimensional anchor* together with the inherited *dimensionless* substrate algebra $\mathfrak{so}(7)$, the Standard-Model and cosmological observables follow at percent-level precision with **zero free continuously-adjustable dimensionless parameters**: the baryon asymmetry $\eta_B = \frac{3}{14}\alpha^4$, the cosmological constant Λ_{cc} , the baryon-to-dark-matter ratio $\Omega_b/\Omega_c = 25\alpha + 75\alpha^2$, the scalar spectral index $n_s = 1 - h_v\alpha$ and tensor ratio $r = 48\alpha^2$, the electroweak vacuum and Higgs mass, gauge-coupling unification at $\alpha_{\text{GUT}}^{-1} = h_v^2$, and a discrete dark-matter mass tower. The framework’s *discrete* structural commitments—the closed form $\alpha = 1/(25\pi\sqrt{3}+1)$, the algebra $\mathfrak{so}(7) = B_3$, the disk curvature $\alpha_{\text{KL}} = 16/25$, and the prefactor integers $(3/14, 8/7, 18, 25)$ —are fixed by the substrate construction rather than fitted to data, but are theoretical choices, not consequences of a deeper dynamical principle established here. The lone dimensional input, M_{Pl} , *cancels from every dimensionless prediction* (verified identically at M_{Pl} and at $10^6 M_{\text{Pl}}$). We give the inheritance mechanism, the derived observable ledger, and the sharp falsifiers— $r = 48\alpha^2 = 2.56 \times 10^{-3}$ ($\sim 5\sigma$ detection by CMB-S4 at design sensitivity, $2\text{--}3\sigma$ discrimination from the Starobinsky benchmark) and a 98 GeV dark-matter state (LZ-G3/DARWIN)—and we demarcate honestly the one quantity that is fixed but not derived (the inflaton-disk radius) and the one that is astrophysical (the event rate). This is the cosmogenic companion to Papers 21a/21b [1, 2], supplying the M_{Pl} -fixing mechanism those papers forward-reference.

1 The cosmogenic bridge and no-hair inheritance

The bridge. A parent black hole driven by accretion to near-extremal spin ($a_{\text{eq}}^* = 0.998052$, the Thorne photon-capture equilibrium [9], which coincides with the substrate K_7 -viability spin) develops an Aretakis instability [12] at its horizon. At saturation a transient **bridge** forms in the *inter-horizon region*: a toroidal T^2 substrate vortex whose toroidal topology is forced from both sides. From the substrate side, K_7 has graph genus 1 (Ringel–Youngs [15]), so its carrier surface must be a torus; from the gravitational side, the Kerr interior supplies exactly one toroidal structure—the region of closed timelike curves around the ring singularity [11]—which is *neither* the ergosurface (topologically S^2) *nor* the near-horizon extremal-Kerr (NHEK) throat’s transverse S^2 [10] where the Aretakis mode operates. The substrate vortex and the Kerr causal torus are *not coincident but co-supported*: the inter-horizon geometric width $r_{\text{gap}} = (r_+ - r_-)/2 = M\sqrt{1 - a^{*2}}$ (a gravitational scale) and the substrate vortex core $r_{\text{core}} = \sqrt{\alpha}M$ (the K_7 coupling scale) are two features of the one bridge, algebraically linked through the impedance match below to the fixed ratio $r_{\text{gap}}/r_{\text{core}} = \sqrt{4\sqrt{2}/9}/(1 + \sqrt{\alpha}) = 0.7304$. Its

Israel junction to the parent fixes the dimensionless product

$$H \times M = h_{\text{Cox}}(\mathfrak{so}(7)) = 6,$$

with the rotational-energy fraction transmitted to the daughter $f_J = (1 - \sqrt{\alpha})^3 \approx 0.765$ —the per-edge Wilson transmission $(1 - \sqrt{\alpha})$ (Paper 17 [3]) over the 3 bridging edges of the polar-vertex Z_3 σ -orbit, the exponent $3 = \text{rank}(\mathfrak{so}(7))$ being the same structural 3 as the η_B numerator and the three generations. The bridge is the locus of a single measurement-like event (the $H^{\otimes 7}$ polarity selection of §3) that commits the daughter’s matter content. The cosmogenic spin is itself substrate-fixed: combining the K_7 -viability ratio $\kappa_{\text{parent}}/\kappa_{\text{daughter}} = 3(1 + \sqrt{\alpha})$ with the Macken impedance $\kappa_{\text{daughter}} = 1/\sqrt{\sqrt{2}\alpha}$ closes to a function of α alone,

$$a_{\text{eq}}^* = \sqrt{1 - \frac{4\sqrt{2}}{9} \frac{\alpha}{(1 + \sqrt{\alpha})^2}} = 0.998052,$$

coincident with the canonical Thorne thin-disk photon-capture equilibrium $a^* = 0.998$ (model-dependent at the $\sim 10^{-3}$ level); the same impedance match pins the ratio of the two bridge radii above.

No-hair inheritance. Just as a Kerr–Newman black hole carries only (M, Q, J) externally regardless of its infall history, a daughter universe carries only a small fixed tuple through inflation, which thermalizes all sub-horizon substrate detail. The inherited set is

$$(M_{\text{Pl}}, \hat{J}_{\text{parent}}, \text{algebra } \mathfrak{so}(7)/Z_3, Z_2 \text{ sign}),$$

mapping onto the three classical Kerr hairs plus the substrate algebra: $J \rightarrow$ the CMB Axis-of-Evil direction; $Q \approx 0 \rightarrow$ charge neutrality; $M \rightarrow$ the Planck scale M_{Pl} ; the order-3 cyclic subgroup $Z_3 \subset G_2 = \text{Aut}(\mathbb{O}) \rightarrow$ three fermion generations and a faint $\ell = 3$ CMB residue; the Z_2 chirality sign $\rightarrow$ matter-over-antimatter (and a cross-arc of sign observables). The action quantum $\hbar$ and the light speed c are likewise inherited as universal unit conventions (identical across daughters; we set $\hbar = c = 1$): c is realized as a/τ_{clock} with the voxel spacing a and clock-tick τ_{clock} themselves set by the M_{Pl} anchor (Paper 21a [1, §9.5]), so it is fixed by M_{Pl} and the substrate, not a separate free input. Exactly one free *dimensional* scale— M_{Pl} —remains.

The M_{Pl} anchor (the load-bearing claim). That anchor is the parent’s **mass-hair**. Because M is a *gravitational* mass, the daughter’s dimensional anchor is its Planck mass M_{Pl} . Every other scale is then M_{Pl} times a dimensionless substrate factor. In particular, the entire fermion and boson spectrum is a set of *derived* rungs of the K_7/K_8 Wilson tower,

$$m = \frac{8}{N_v} \alpha^{N_e/2} (1 + \alpha/7)(1 + 3\alpha^2) M_{\text{Pl}},$$

with the electron the $(N_e = 21, N_v = 7)$ rung, reproducing m_e to -5.5 ppm at NNLO (Paper 17 [3]). The electroweak vacuum is $v_{\text{EW}} = (h_v/2) \alpha^8 M_{\text{Pl}} = 245.5$ GeV (-0.3%). The electron mass is thus *predicted from* M_{Pl} , not an input.

One scale, zero dimensionless parameters. Since every scale is $M_{\text{Pl}} \times (\alpha\text{-power})$, M_{Pl} **cancels from every dimensionless observable**— m_e/M_{Pl} , $v_{\text{EW}}/M_{\text{Pl}}$, $\Lambda_{\text{cc}}/M_{\text{Pl}}^4$, m_H/v_{EW} , Ω_b/Ω_c , η_B , n_s , r —which we verify identically at M_{Pl} and at $10^6 M_{\text{Pl}}$. The daughter therefore has **one free scale and zero free dimensionless parameters**. The role of M_{Pl} (the anchor) is inherited; its numerical value (which parent) is contingent and, because it cancels from all ratios, unobservable in any dimensionless

test. This is the precise content of the loose “one free parameter” statement of earlier work. The count is of *continuously-adjustable* parameters: the framework still makes discrete structural commitments—the closed form of α , the algebra $\mathfrak{so}(7) = B_3$, and the disk curvature $\alpha_{\text{KL}} = 16/25$ —which are fixed by the construction rather than fitted to data, but are choices of theory, not consequences of a deeper dynamical principle established here.

2 Substrate primitives

The framework rests on a single dimensionless constant, the trefoil-residue fine-structure value

$$\alpha = \frac{1}{25\pi\sqrt{3} + 1} = 7.2974 \times 10^{-3},$$

equal to the Thomson-limit α_{em} to 7.6 ppm (a closed-form leading-order value, not an exactness claim). The substrate algebra is $\mathfrak{so}(7) = B_3$ (rank 3, dual Coxeter $h_v = 5$ (the same substrate invariant as Paper 21a’s K_7 Hopf-vacuum closure-2 vertex count [1, §2], since K_7 ’s 21 edges = $\dim \mathfrak{so}(7)$), Coxeter $h_{\text{Cox}} = 6$, $\dim 21$, $N_{\text{pos}} = 9$ positive roots). Its physical carrier is the complete graph K_7 , identified with the Steane $[[7, 1, 3]]$ code, the Fano plane, and the octonion imaginaries—one seven-point object—an identification whose Steane-code realization reproduces the predicted stabilizer syndromes on IBM Heron r2 and across architectures (Paper 21b [2]). As Paper 21b emphasizes, this is evidence that the walk-to-Pauli mapping is correctly implemented and noise-robust on hardware, *not* independent confirmation that the Standard Model *is* encoded as K_7 walks; the identification remains a theoretical ansatz. Two integers thread the whole observable set: the “DNA” pair $18 = \text{rank} \cdot h_{\text{Cox}}$ and $25 = h_v^2$.

3 Matter genesis: $\eta_B = \frac{3}{14}\alpha^4$

The $H^{\otimes 7}$ polarity-selection event at the bridge (the measurement-like commitment of §1) is what fixes the daughter’s matter content. Under the $K_7 \equiv \text{Steane } [[7, 1, 3]]$ identification (§2; Paper 21a [1]), matter and antimatter are not two condensate species but the two readout-basis selections of a single self-dual code, exchanged by the transversal Hadamard $H^{\otimes 7}$ —the substrate CPT operation. The cosmogenic asymmetry is the bias of that basis selection:

$$\eta_B = \frac{\text{rank}(\mathfrak{so}(7))}{\dim(G_2)} \alpha^4 = \frac{3}{14} \alpha^4 = 6.08 \times 10^{-10},$$

agreeing with the Planck/BBN value $(6.10 \pm 0.10) \times 10^{-10}$ at 0.38%, well inside the 1.6% experimental error, with $\text{sgn}(\eta_B) = \text{sgn}(J_{\text{parent}})$ —the matter-over-antimatter sign is the inherited Z_2 chirality (§1).

The prefactor is structural, not fitted. The numerator $3 = \text{rank}(\mathfrak{so}(7))$ is the number of fermion generations, which contribute additively to the CP-asymmetric amplitude; the denominator $14 = \dim(G_2) = \dim \text{Aut}(\mathbb{O})$ is the substrate-automorphism phase space over which that amplitude is distributed. The power α^4 is NLO²: two NR-Hamilton CP-violating vertices, each carrying an α^2 loop suppression. The α^4 scaling is tightly constrained: α^3 over-predicts η_B by $\sim 10^4$ while QED-loop-natural prefactors ($\alpha^4/4\pi$, $\alpha^4/16\pi$) under-predict by 60–95%, so the substrate (non- π) prefactor $3/14$ at α^4 is forced rather than fitted.

Two presentations of one mechanism. The result admits two formal presentations converging on identical numerics by construction: a *bridge-event* reading, in which η_B is the $H^{\otimes 7}$ stabilizer-basis-selection bias at the cosmogenic phase transition; and a *carrier-knot* reading, in which the asymmetry

is a walk-population imbalance sourced by the NR-Hamilton ($d = 3$, non-quadratic-residue Hamilton-cycle) substructure of K_7 walks (the same Law-2 Hamilton bifurcation that fixes the mass-formula sectors, Paper 21a [1], §4). These are dual descriptions of the same QR/NR substrate asymmetry—operator-side and state-side, related by the Pauli-bit-level QR/NR $\leftrightarrow X/Z$ duality (Paper 21a [1], §5.3)—*not* independent derivations; a genuinely independent route via the K_8 sector (e.g. the Paper 20 [6] neutrino CP phase) would constitute substantive cross-confirmation and is the natural follow-up. The CP-violating channel is distinct from the CKM Jarlskog invariant, which alone gives $\eta_B \sim 10^{-20}$ at $\alpha^{5/2}$; the substrate channel operates ten orders of magnitude higher, at α^4 , matching observation without beyond-Standard-Model fields. All three Sakharov conditions are met structurally: baryon-number violation by the discrete change of walk-homology (p, q) under transduction, CP-violation by the NR-Hamilton phase, and the out-of-equilibrium condition by the cosmogenic bridge itself (§1). The $K_7 \equiv$ Steane identification underwriting both routes is hardware-anchored (§2; Paper 21b [2]).

4 Cosmological partition: Ω_b/Ω_c and Λ_{cc}

The bridge condensate partitions between a K_7 -Steane-active (baryonic) channel and a K_8 -extension-only (cold dark matter) channel. The leading Wilson-amplitude ratio gives

$$\Omega_b/\Omega_c = 25\alpha + 75\alpha^2 = 0.1864,$$

within 60 ppm of the Planck 2018 ratio $\Omega_b h^2/\Omega_c h^2 = 0.02237/0.1200 = 0.18642$ [14], where $25 = q_{\text{cinq}}^2$ (the cinquefoil determinant) at leading order and $75 = 3 \times 25$ (three generations) at next-to-leading order. The cosmological constant is set by the same anchor and algebra,

$$\Lambda_{cc}/M_{\text{Pl}}^4 = (m_e/M_{\text{Pl}})^4 \alpha^{16} h_{\text{Cox}},$$

reproducing the observed dark-energy density to three digits—and, since m_e/M_{Pl} is itself an α -power (§1), this is a pure substrate number times M_{Pl}^4 .

5 Inflation: an α -attractor forced by the bridge

The type is a theorem. The inflaton is a modulus of the substrate surface—the K_7 Heffter torus T^2 , which becomes the genus-2 surface Σ_2 at the cosmogenic topology change $T^2 \# T^2 = \Sigma_2$. The Teichmüller space of the torus *is* the Poincaré disk $SL(2, \mathbb{R})/SO(2)$ with the constant-curvature Weil–Petersson metric, so the inflaton field space is a hyperbolic disk and inflation is an **α -attractor** [13] as a theorem of Riemann-surface moduli, not an ad-hoc potential choice. (The same $SL(2, \mathbb{R})$ is the NHEK bridge isometry.) Genus-2 carries three moduli, matching the three substrate dynamical fields (inflaton, Higgs, Z_2).

The observables. Equating the α -attractor relation $n_s = 1 - 2/N_e$ with the substrate value $n_s = 1 - h_v \alpha$ fixes the e-fold count in closed form, $N_e = 2/(h_v \alpha) = 10\pi\sqrt{3} + 2/5 \approx 54.81$ (an independent $V_0 = \bar{\lambda}_C^3$ bookkeeping route agrees at 2.7%); the α -attractor relations then give

$$\begin{aligned} n_s &= 1 - 2/N_e = 1 - h_v \alpha = 0.9635 \quad (\text{Planck } 0.9649 \pm 0.0042), \\ r &= 12 \alpha_{\text{KL}}/N_e^2 = 48\alpha^2 = 2.56 \times 10^{-3}, \end{aligned}$$

where the single shape parameter is the disk curvature $\alpha_{\text{KL}} = ((\text{rank} + 1)/h_v)^2 = 16/25$ (the Kähler-manifold curvature of the α -attractor, a distinct quantity from the fine-structure α of §2; the B_3 substrate fixes both). **Honest demarcation:** α_{KL} is *fixed* by the B_3 substrate with zero free

parameters—the α -attractor *type* is derived (above), and n_s , r , and the Kähler curvature $R_K = -25/24$ all follow—but the disk *radius* $\sqrt{\alpha_{\text{KL}}} = 4/5$ is an **irreducible substrate datum**: seven independent derivation routes (coherent-state, spinor coadjoint, Potts c^2 , multi-node, gauged-WZW, JT/Schwarzian, and the sugra/moduli reduction) are exhausted, unified by the observation that each supplies a single linear/integer handle while α_{KL} is a ratio of two squared affine integers. The prediction $r = 48\alpha^2 = 2.56 \times 10^{-3}$ is falsifiable regardless of the radius’s origin: at the CMB-S4 design sensitivity $\sigma(r) = 5 \times 10^{-4}$ it is a $\sim 5\sigma$ *detection* (matching CMB-S4’s $r > 0.003$ target), with LiteBIRD ($\delta r \lesssim 10^{-3}$) an independent probe. The Starobinsky limit of the same α -attractor family ($\alpha_{\text{KL}} = 1$) gives $r = 12/N_e^2 = 75\alpha^2$ at the substrate N_e , so the substrate–Starobinsky separation is $\Delta r = 27\alpha^2 = 1.4 \times 10^{-3}$, i.e. $2\text{--}3\sigma$ at design sensitivity; $> 5\sigma$ discrimination requires post-CMB-S4 (PICO/PIXIE-class) instruments.

6 Dark and electroweak sectors: the K_8 tower

The Higgs sector follows from the M_{Pl} anchor: $v_{\text{EW}} = (h_v/2) \alpha^8 M_{\text{Pl}} = 245.5 \text{ GeV}$ (-0.3%), quartic $\lambda_H = 2N_{\text{pos}} \alpha = 18\alpha$, and $m_H = \sqrt{2\lambda_H} v_{\text{EW}} = 125.8 \text{ GeV}$ ($+0.4\%$)—the same DNA integer $18 = \text{rank} \cdot h_{\text{Cox}}$ appearing in λ_H and in H_0 . Gauge couplings unify at $\alpha_{\text{GUT}}^{-1} = h_v^2 = 25$ (MSSM range 24–26) without low-energy supersymmetry: the K_8 mass tower’s dark-matter states play the role of the threshold corrections.

The K_8 edge-orbit tower yields a discrete multi-species dark sector—a 38 keV warm state, a 98 **GeV WIMP** ($N_e = 16$, i.e. $\alpha^8 M_{\text{Pl}}$), and a 1.15 TeV heavy state—the WIMP being the most testable substrate prediction. Dark matter is described by the K_8 tower alone (this supersedes earlier condensate-species treatments).

7 Falsifiable predictions

1. $r = 48\alpha^2 = 2.56 \times 10^{-3}$ —a single point in the (n_s, r) plane at $(0.9635, 2.56 \times 10^{-3})$. First-decade falsifier: CMB-S4 *detects* this value at $\sim 5\sigma$ at design sensitivity $\sigma(r) = 5 \times 10^{-4}$, with LiteBIRD independent confirmation; discrimination from the Starobinsky limit ($r = 75\alpha^2$) is $2\text{--}3\sigma$, with higher- σ discrimination requiring post-CMB-S4 instruments.
2. **98 GeV K_8 WIMP**—substrate-fixed at $\alpha^8 M_{\text{Pl}}$; LZ-G3 (2026–28) probes most of the σ_{SI} bracket, DARWIN/XLZD (2030+) the whole of it.
3. **Multi-species DM tower** (38 keV + 98 GeV + 1.15 TeV)—a distinguishing signature versus single-component WIMP/axion scenarios.
4. Σm_ν **tower**—the active-neutrino rung ($N_e = 28$) gives a DESI “clock” falsifier (Paper 20 [6]).
5. **No low-energy SUSY**—the substrate predicts MSSM-like unification *without* superpartners. Current LHC Run-3 null results on natural SUSY are consistent with this; HL-LHC will probe substantially more parameter space, and an LHC SUSY discovery would falsify the K_8 -as-thresholds picture.
6. **Cosmogenic-parent AGN population**—the substrate fixes the parent *selection* (mass band $M^* \cdot [1/\sqrt{2}, \sqrt{2}]$, spin $a^* \approx 0.998$), predicting a candidate density $n_{\text{cand}} \approx 3.3 \times 10^{-6} \text{ Mpc}^{-3}$ ($\sim 10^3$ within $z < 0.1$). This is a population-level prediction, not a parent identification: no present observable distinguishes “the” parent from generic high-spin AGN, so we include it as a structural consequence rather than a near-term falsifier (sharpening needs a population

statistic—spin distribution or mass-band fraction—separating substrate-canonical parents from generic populations).

7. **Axis of Evil**—the inherited spin axis, correlated with the cross-arc Z_2 sign. We treat this as suggestive rather than decisive: the statistical significance of the CMB low- ℓ alignments remains debated.

8 Discussion

Versus Cosmological Natural Selection. Like Smolin’s CNS [8], universes are born inside black holes; unlike CNS, the daughter’s *dimensionless* constants are **substrate-fixed and do not drift**: they are set by the inherited $\mathfrak{so}(7)$ algebra rather than by the parent’s particular history, so daughters reproduce them up to the usual renormalization-group running and quantum corrections, not by Darwinian tuning. There is no selection pressure on the dimensionless constants; the genealogy is *algebraic*, not evolutionary. The cousins-per-parent number is $O(1)$ (shallow genealogy), so cousin-clustering does not lift the no-hair degeneracy.

What is and is not derived. The framework derives > 40 dimensionless observables at percent precision from α and the inherited algebra, with one free scale (M_{Pl}) that cancels from all of them. Three honest limits: (i) the inflaton-disk *radius* $\sqrt{\alpha_{\text{KL}}} = 4/5$ is fixed by the B_3 substrate but not reducible to a more primitive dynamical quantity (§5)—the remaining derivation avenue is a full string compactification computing the Weil–Petersson $\mathfrak{so}(7)$ -dressing from substrate fluxes; (ii) the cosmogenic *event rate* normalization is astrophysical, though its *selection* (parent mass + spin) is substrate-fixed (§7); (iii) the *bridge* is an *assumed* saturation endpoint of the Aretakis instability—a linear-level theorem on extremal horizons whose nonlinear endpoint we posit (not derive) to be the substrate-mediating T^2 vortex, with a controlled extremal-Kerr saturation analysis the key missing dynamical step.

9 Observable ledger

All entries are M_{Pl} -independent (dimensionless, or ratios to M_{Pl}); one free scale, zero free dimensionless parameters.

Observable	Substrate form	Predicted	Observed	Δ
η_B	$\frac{3}{14}\alpha^4$	6.08×10^{-10}	6.10×10^{-10}	0.38%
Ω_b/Ω_c	$25\alpha + 75\alpha^2$	0.1864	0.1864	60 ppm
n_s	$1 - h_v\alpha$	0.9635	0.9649 ± 0.0042	0.3σ
r	$48\alpha^2$	2.56×10^{-3}	< 0.036	(target)
a_{eq}^*	$\sqrt{1 - \frac{4\sqrt{2}}{9}\alpha/(1 + \sqrt{\alpha})^2}$	0.998052	≈ 0.998 (Thorne)	within model unc.
m_e	$(8/7)\alpha^{21/2}(\text{NNLO}) M_{\text{Pl}}$	0.5110 MeV	0.5110 MeV	−5.5 ppm
v_{EW}	$(h_v/2)\alpha^8 M_{\text{Pl}}$	245.5 GeV	246.22 GeV	−0.3%
m_H	$\sqrt{2} \cdot 18\alpha v_{\text{EW}}$	125.8 GeV	125.25 GeV	+0.4%
α_{GUT}^{-1}	h_v^2	25	24–26 (MSSM)	in range
$\Lambda_{\text{cc}}/M_{\text{Pl}}^4$	$(m_e/M_{\text{Pl}})^4 \alpha^{16} h_{\text{Cox}}$	1.19×10^{-123}	1.2×10^{-123}	−0.7%
M_{WIMP}	$\alpha^8 M_{\text{Pl}} (N_e = 16)$	98.2 GeV	(LZ-G3 target)	—

Data and code availability

The cosmogenesis and inheritance computations—the M_{Pl} -anchor cancellation check (`cosmogenesis_M_Pl_anchor_inheritance.py`), the NHEK bridge setup, the α -attractor radius routes, the η_B candidate sweep (`baryon_asymmetry_memo.md`, `nwt_baryogenesis_chirality.py`), and the cosmogenic-AGN rate density—are available in the public analyses repositories at <https://github.com/JimGalasyn/null-worldtube> and <https://github.com/JimGalasyn/vortex-vision>. The substrate-algebraic computation library underlying the α , $\mathfrak{so}(7)$, and K_8 identities used throughout is `nwt-substrate` v0.2.0 (Zenodo DOI 10.5281/zenodo.20398451; concept DOI 10.5281/zenodo.20012027 for all versions), importing the canonical constants from `isa.constants`.

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